SUPERSTORE

GLOBAL SALES ANALYTICS

SALES & PROFITABILITY ANALYSIS

An Interactive Business Intelligence Report
using Microsoft Power BI

2011 – 2014



Power BI

1. PROJECT OVERVIEW

This report presents an analysis of the Superstore Global Sales dataset using the Power BI Analysis Dashboard. The dataset contains customer orders, shipping information, product categories, sales, discounts, profit and shipping costs. The dashboard expands the earlier analysis with a consolidated executive view covering financial performance, customer segments, product categories, geographic markets, shipping modes and time-based sales trends.

The analysis covers the complete period from 2011 to 2014. The Executive Dashboard reports total sales of $12.64M, total profit of $1.47M, approximately 25.0K orders, 795 customers, 178.3K units sold and an average shipping cost of 26.38.



2. INTRODUCTION

The objective of this project is to explore Superstore Global sales data and identify trends and patterns that can support strategic business decisions. The Power BI dashboard focuses on sales growth, profitability, customer segmentation, product performance, geographic contribution, shipping behavior and seasonal demand.

The analysis helps answer key business questions:

- Which categories and sub-categories generate the most value?
- Which customer segments contribute most to sales?
- Which countries and markets are strongest?
- How have sales and profit changed over time?
- Which shipping modes dominate?
- where are discounts creating profitability risk?

3. IMPORTANCE OF DATA ANALYSIS

Data analysis is important for Superstore because revenue growth alone does not guarantee profitable growth. The dashboard makes it possible to connect sales volume with profit, customer mix, product mix and operating cost.

- Customer Insights help identify the highest-value segments and support targeted marketing.
- Product and Profitability Analysis helps prioritize high-performing sub-categories and identify loss-making products.
- Geographic Analysis helps distinguish major revenue markets from smaller but potentially attractive markets.
- Shipping and Order Analysis supports logistics planning and cost-efficiency decisions.
- Seasonal Trend Analysis helps prepare inventory, promotions and fulfillment capacity for peak demand.
- Discount Analysis highlights where aggressive pricing is reducing or eliminating profit.

4. DATA PREPARATION

The project included data-cleaning and preprocessing steps before analysis. The report retains that analytical foundation while using the Power BI dashboard for the business findings.

4.1 DATA CLEANING AND PREPROCESSING

- Duplicate records were checked in the original project.
- Date fields such as order date and ship date were prepared for time-series analysis.
- Sales was converted to a numeric field for aggregation and analysis.
- Categorical fields including category, sub-category, segment and ship mode were standardized for consistent reporting.
- The dashboard covers the full 2011–2014 period and provides interactive filters for year, segment, ship mode, order priority, market, order date, category and country.

The dashboard indicates approximately 51,290 order lines and coverage across 147 countries.

5. PROJECT ANALYSIS

5.1 EXPLORATORY DATA ANALYSIS (EDA)

The EDA examines descriptive performance measures and business dimensions including sales, profit, orders, customers, units sold, shipping cost, category, sub-category, segment, country, market and time.

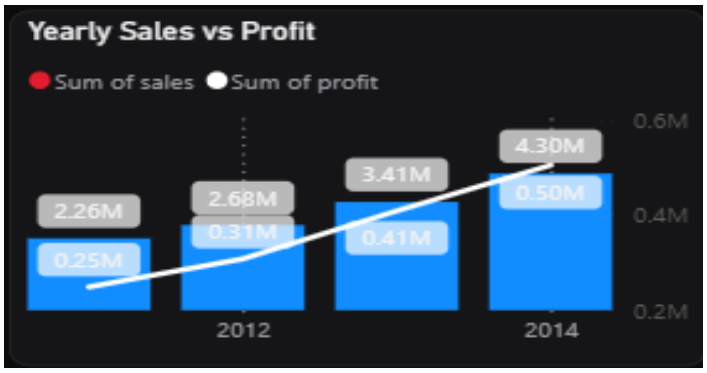
Measure	Dashboard value
Total Sales	$12.64M
Total Profit	$1.47M
Total Orders	25.0K
Total Customers	795
Units Sold	178.3K
Average Shipping Cost	26.38

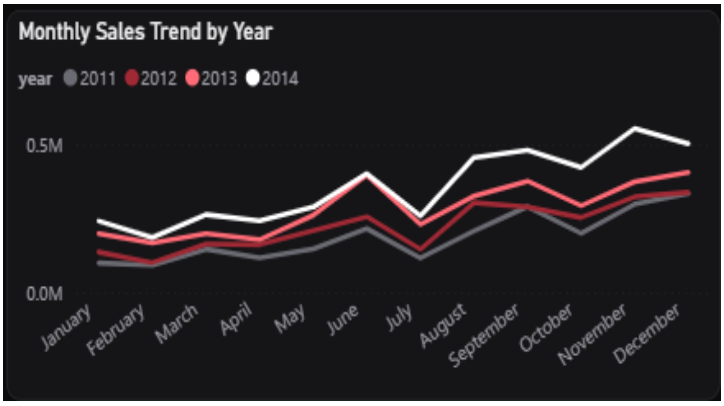
5.2 SALES AND PROFIT TRENDS

The annual trend shows sustained growth across all four years. Sales increased from $2.26M in 2011 to $4.30M in 2014, while profit increased from $0.25M to $0.50M.

Year	Sales	Profit
2011	$2.26M	$0.25M
2012	$2.68M	$0.31M
2013	$3.41M	$0.41M
2014	$4.30M	$0.50M

Overall sales growth from 2011 to 2014 is approximately 90%. The dashboard's Key Insights panel also highlights that 2014 added approximately 26% sales growth compared with 2013.

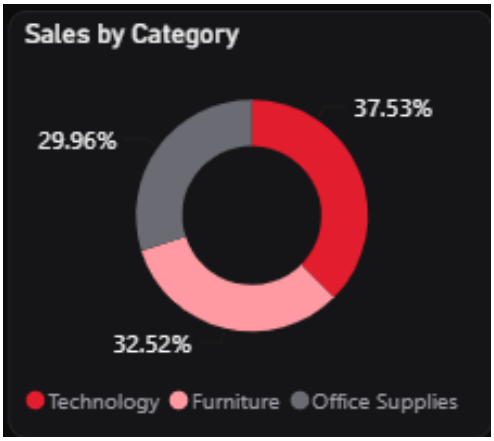




5.3 SALES BY CATEGORY

Technology is the largest sales category at 37.53%, followed by Furniture at 32.52% and Office Supplies at 29.96%. The category mix is relatively balanced, but the profitability profile differs substantially.

Category	Sales share	Dashboard profitability observation
Technology	37.53%	Approximately 14.0% margin
Furniture	32.52%	Approximately 7.0% margin
Office Supplies	29.96%	Large sales contribution; monitor margin with volume

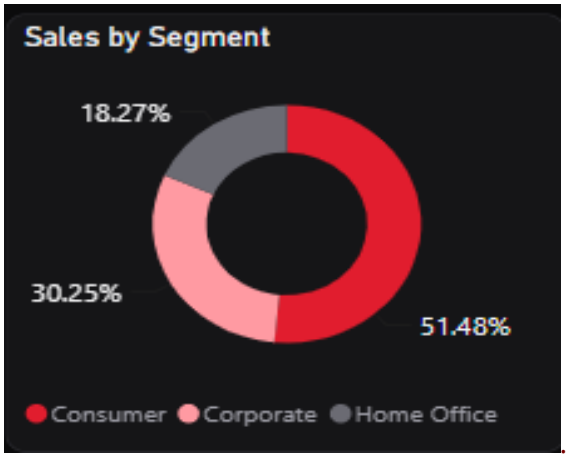


5.4 CUSTOMER SEGMENT ANALYSIS

Consumer customers contribute the largest share of sales at 51.48%, followed by Corporate at 30.25% and Home Office at 18.27%.

Segment	Sales share
Consumer	51.48%
Corporate	30.25%
Home Office	18.27%

Consumer therefore remains the primary revenue segment. Corporate and Home Office are smaller but meaningful segments that can provide opportunities for targeted growth.

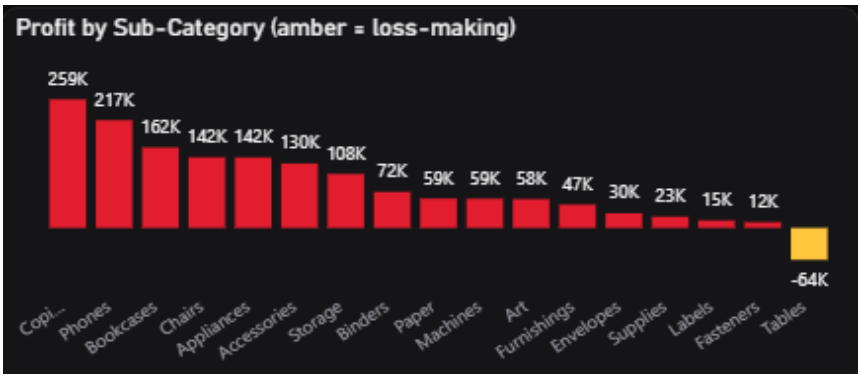


5.5 PRODUCT AND SUB-CATEGORY PROFITABILITY

The profitability view shows strong performance in Copiers and Phones, while Tables is the clear loss-making sub-category.

Sub-category	Profit shown
Copiers	$259K
Phones	$217K
Bookcases	$162K
Chairs	$142K
Appliances	$142K
Accessories	$130K
Storage	$108K
Binders	$72K
Paper	$59K
Machines	$59K
Art	$58K
Furnishings	$58K
Envelopes	$47K
Supplies	$30K
Labels	$23K
Fasteners	$15K
Tables	-$64K

The Tables sub-category records approximately $64K in losses. The dashboard's Key Insights panel further notes that Tables have an average discount of approximately 29%, while orders with discounts above 30% run at approximately a -46.8% margin.

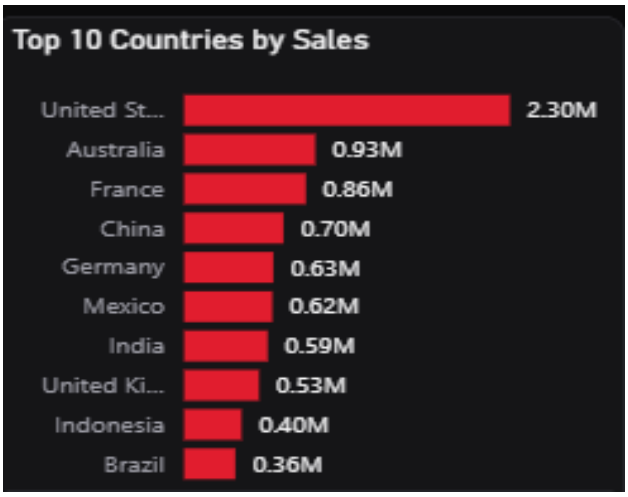


5.6 GEOGRAPHIC ANALYSIS

The dashboard's Top 10 Countries by Sales visual shows the United States as the leading country, followed by Australia and France.

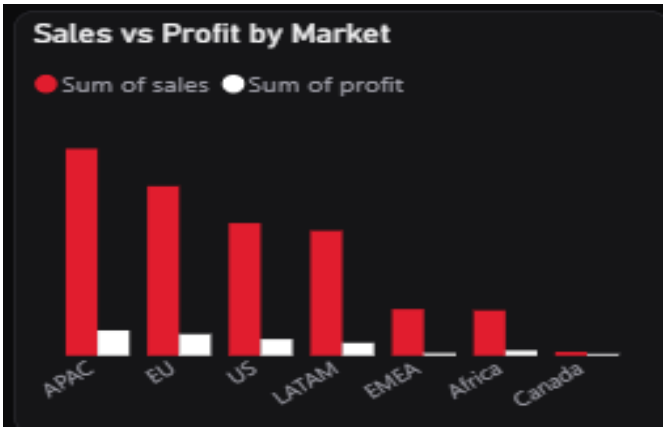
Country	Sales shown
United States	$2.30M
Australia	$0.93M
France	$0.86M
China	$0.70M
Germany	$0.63M
Mexico	$0.62M
India	$0.59M
United Kingdom	$0.53M

The geographic results demonstrate that the business is internationally diversified, with substantial contributions across North America, Europe, Asia and Australia.



5.7 MARKET ANALYSIS

The Sales vs Profit by Market visual compares APAC, EU, US, LATAM, EMEA, Africa and Canada. The key purpose of this view is to compare revenue scale with profit contribution, helping identify markets where sales growth is or is not translating into healthy profitability.

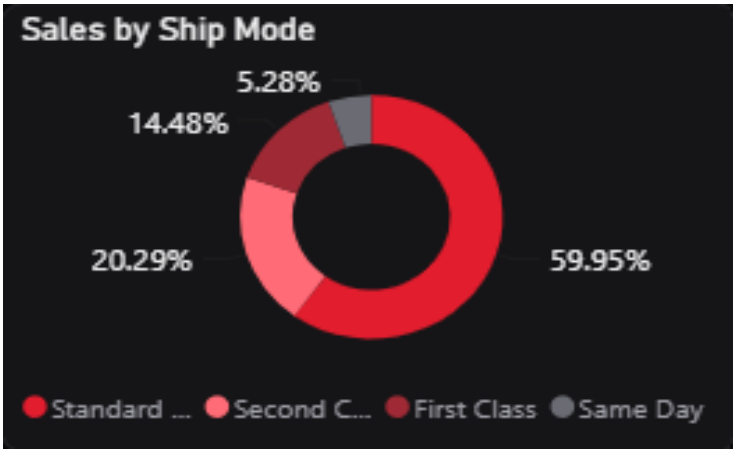


5.8 SHIPPING MODE ANALYSIS

Standard Class is the dominant shipping mode at 59.95%, followed by Second Class at 20.29%, First Class at 14.48% and Same Day at 5.28%.

Ship Mode	Share shown
Standard Class	59.95%
Second Class	20.29%
First Class	14.48%
Same Day	5.28%

The dominance of Standard Class indicates that customers generally use the cost-effective shipping option. The reported average shipping cost of 26.38 should therefore be monitored alongside service-level requirements.



6. POWER BI INSIGHTS SUMMARY

The Executive Dashboard produces the following major insights:

- Sales grew by approximately 90% from 2011 to 2014, increasing from $2.26M to $4.30M.
- 2014 recorded the highest annual sales at $4.30M and approximately $0.50M in profit.
- Technology is the largest category at 37.53% of sales and has an approximately 14.0% margin.
- Furniture contributes 32.52% of sales but has an approximately 7.0% margin, indicating weaker profitability relative to its sales contribution.
- Consumer is the largest customer segment at 51.48% of sales.
- Copiers and Phones are the strongest profit-generating sub-categories shown in the dashboard.
- Tables is loss-making at approximately -$64K and is associated with high discounting.
- The dashboard highlights that orders with discounts above 30% run at approximately a -46.8% margin.
- The United States is the largest visible country market at approximately $2.30M in sales.
- Standard Class represents 59.95% of the shipping mix.
- Monthly sales generally strengthen toward the later part of the year, with the year-end period showing stronger demand.



7. CONCLUSION AND RECOMMENDATIONS

The analysis shows a business with strong revenue momentum and a broad international customer base. Sales nearly doubled from 2011 to 2014, and both sales and profit reached their highest levels in 2014. However, the dashboard also shows that revenue growth must be managed carefully to protect margins.

RECOMMENDATIONS

- Product Focus: Continue promoting high-value and profitable Technology products, with particular attention to strong sub-categories such as Copiers and Phones.
- Discount Control: Establish tighter controls for discounts above 30%, because the dashboard indicates a severe negative margin at this level.
- Tables Review: Investigate the approximately $64K Tables loss at SKU/product level and review pricing, discounts, freight and product costs.
- Furniture Profitability: Improve the approximately 7.0% Furniture margin through product-mix, pricing and discount optimization.
- Customer Segmentation Strategy: Maintain strong Consumer targeting while developing focused offers for Corporate and Home Office customers.
- Shipping Optimization: Keep Standard Class as the core shipping option while monitoring the economics of faster delivery modes.
- Seasonal Preparation: Increase inventory, marketing and fulfillment readiness before the stronger year-end sales period.
- Geographic and Market Optimization: Evaluate markets using both sales and profit rather than sales alone.